ALFRED LEIGH

Aspiring Aerospace Engineer

Theydon Bois, Essex • +44 7940 464999 • alfxzmail@gmail.com • alfred-leigh.co.uk • github.com/a365l

PROFILE

Year 12 student (A-level Mathematics, Further Mathematics, Physics and Computer Science) committed to a career in aerospace engineering. I design and build real hardware from first principles: currently a from-scratch quadcopter and a 72V electric enduro motorcycle, each documented to industry conventions (product design specifications, trade studies, engineering calculations, failure logs) and public on GitHub. Completed a week of civil engineering work experience with Berkeley Group on live London sites in summer 2026. Seeking aerospace placements and competition opportunities.

ENGINEERING PROJECTS

First-Principles Quadcopter · github.com/a365l/quadcopter-project — Jun 2026 – present

- Wrote a Product Design Specification with 15 quantified requirements (3:1 thrust-to-weight, arm resonance ≥ 1.5× motor frequency, ≤3.5 kg all-up weight) before touching CAD.
- Building a first-principles calculation set (momentum-theory propulsion sizing, Euler-Bernoulli cantilever resonance, bolt preload), each to be validated against thrust-stand and flight-test measurements.
- Pressure-tested feasibility with researchers at Imperial College London; £700 budget, running to first flight alongside sixth form.

72V Enduro Electric Motorcycle · github.com/a365l/enduro-emotorcycle-build — 2025 – present

- Built a full electric enduro motorcycle from a bare frame across three iterations, culminating in a QS205 hub motor and Fardriver ND72450 controller (rated 450 A peak, tuned to 100 A battery / 300 A phase to match the pack); every phase photo-documented on GitHub.
- Predicted ~55 mph top speed and ~3.6 s 0–30 mph from first principles; maintain a failure log (symptom → root cause → fix → retest) and test-vs-prediction records for every subsystem.
- Diagnosed a dead-on-arrival controller (shorted MOSFETs) with a 10-second continuity check; designed a custom throttle adapter harness; ~£3,000 build part-funded by stripping and flipping donor bikes.

Roboscan · RF security multi-tool: sub-GHz/NFC/IR interface device (CC1101, ESP32/Pi Pico) plus network-enumeration tooling; first major embedded build.

WORK EXPERIENCE

Berkeley Group · civil engineering placement, Bermondsey Place & Trent Park, London — Jun–Jul 2026

- One-week placement across two live sites, shadowing site engineers on temporary works, substructure, piling and setting out, and working alongside quantity surveyors, planners and subcontractors.
- Hands-on with a Leica total station and rotating laser level for setting out and level checks; observed concrete QC catch a failed slump test on a live pour, alongside 7- and 28-day cube testing.
- Worked through manhole formation-level calculations and BS 8666 rebar shape codes on site; wrote the week up as a 20-slide technical report.

Kras Carpentry · paid commercial work — 2024 – present

- Hands-on carpentry with power tools, basic household plumbing and electrical installation; designed, built and maintain the company website, krascarpentry.co.uk.

EDUCATION

Sixth Form A-levels (from Sept 2026): Mathematics, Further Mathematics, Physics, Computer Science — 2026 – 2028

Debden Park High School · 9 GCSEs, 8 at grade 7 or above: Mathematics 8, Physics 8, Chemistry 8, Computer Science 8, Biology 7, Geography 7, English Literature 7, English Language 7 (Spoken Language: Merit), Physical Education 6 — 2021 – 2026

SKILLS

- **Engineering practice:** product design specifications, trade studies, first-principles calculation, failure logging, test-vs-prediction validation, BOM & budget management
- **Electronics:** soldering & micro-soldering, crimped vehicle looms, 72V high-voltage battery systems, BLDC motor control (FOC), embedded systems (ESP32, Pi Pico, Raspberry Pi)
- **Software:** C++, Python, Kotlin, JavaScript/React, Linux (Debian/Arch) administration & networking; practical cyber-security (Kali, TryHackMe)

LANGUAGES & INTERESTS

English (native), Slovak (fluent), Czech (fluent) • Sailing (RYA Level 4) • Bouldering (V6) • Boxing & Brazilian Jiu-Jitsu • Competitive swimming • Guitar (8 years)